

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



**777**

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

---

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1 Product identifier

Trade name : 777

Sales Number : 6203372

### 1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub-stance/Mixture : Fragrance for consumer product

### 1.3 Details of the supplier of the safety data sheet

Company : IFF (NEDERLAND) BV  
ZEVENHEUVELENWEG 60  
5048 AN TILBURG

Telephone : +31134642211

Telefax : +31134636032

E-mail address of person responsible for the SDS : sds@iff.com

### 1.4 Emergency telephone number

ORFILA (INRS): + 33 (0)1 45 42 59 59

---

## SECTION 2: Hazards identification

### 2.1 Classification of the substance or mixture

#### Classification (REGULATION (EC) No 1272/2008)

|                                                |                                                        |
|------------------------------------------------|--------------------------------------------------------|
| Skin irritation, Category 2                    | H315: Causes skin irritation.                          |
| Eye irritation, Category 2                     | H319: Causes serious eye irritation.                   |
| Skin sensitisation, Category 1                 | H317: May cause an allergic skin reaction.             |
| Long-term (chronic) aquatic hazard, Category 2 | H411: Toxic to aquatic life with long lasting effects. |

### 2.2 Label elements

#### Labelling (REGULATION (EC) No 1272/2008)

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

**777**

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

Hazard pictograms :



Signal word : Warning

Hazard statements :

- H315 Causes skin irritation.
- H317 May cause an allergic skin reaction.
- H319 Causes serious eye irritation.
- H411 Toxic to aquatic life with long lasting effects.

Precautionary statements :

**Prevention:**

- P261 Avoid breathing mist or vapours.
- P264 Wash skin thoroughly after handling.
- P280 Wear protective gloves/ eye protection/ face protection.

**Response:**

- P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
- P337 + P313 If eye irritation persists: Get medical advice/ attention.
- P391 Collect spillage.

## Hazardous components which must be listed on the label:

|                                                         |                                         |
|---------------------------------------------------------|-----------------------------------------|
| 68155-66-8,<br>54464-57-2,<br>68155-67-9,<br>54464-59-4 | Tetramethyl Acetyloctahydronaphthalenes |
| 33704-61-9                                              | Dihydro pentamethylindanone             |
| 127-51-5                                                | Alpha-Isomethyl Ionone                  |
| 106-22-9,<br>7540-51-4                                  | Citronellol                             |
| 106-24-1                                                | geraniol                                |
| 1205-17-0                                               | METHYLENEDIOXYPHENYL METHYLPROPANAL     |
| 87-44-5                                                 | Beta-Caryophyllene                      |
| 80-56-8, 7785-<br>26-4                                  | pin-2(3)-ene                            |
| 5989-27-5                                               | (R)-p-mentha-1,8-diene                  |
| 127-91-3                                                | BETA-PINENES                            |
| 13466-78-9                                              | 3,7,7-trimethylbicyclo[4.1.0]hept-3-ene |
| 23696-85-7                                              | ROSE KETONE-4                           |
| 116-26-7                                                | 2,3-dihydro-2,2,6-trimethylbenzaldehyde |

## 2.3 Other hazards

None reasonably foreseeable.

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Ecological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

777

Version 1.0      Revision Date: 20.07.2023      SDS Number: 300001111100      Date of last issue: -  
Date of first issue: 20.07.2023

Toxicological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

## SECTION 3: Composition/information on ingredients

### 3.2 Mixtures

#### Components

| Chemical name                                                                                                                       | CAS-No.<br>EC-No.<br>Index-No.<br>Registration number                                                  |                                                                                                                           | Concentration<br>(% w/w) |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one                                                             | 68155-66-8<br>915-730-3<br>01-2119489989-04                                                            | Skin Irrit. 2; H315<br>Skin Sens. 1B; H317<br>Aquatic Chronic 2;<br>H411                                                  | >= 30 - < 50             |
| (2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol                                                                    | 28219-61-6<br>701-122-3<br>01-2119529224-45                                                            | Eye Irrit. 2; H319<br>Aquatic Chronic 2;<br>H411<br>Skin Irrit. 2; H315<br><br>M-Factor (Acute aquatic toxicity): 1       | >= 10 - < 20             |
| 1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one                                                                          | 33704-61-9<br>251-649-3<br>01-2119977131-40                                                            | Aquatic Chronic 2;<br>H411<br>Skin Sens. 1B; H317<br>Eye Irrit. 2; H319<br>Skin Irrit. 2; H315                            | >= 2,5 - < 10            |
| (+/-) trans-3,3-dimethyl-5-(2,2,3-trimethyl-cyclopent-3-en-1-yl)pent-4-en-2-ol                                                      | 107898-54-4<br>411-580-3<br>603-150-00-0<br>01-0000015895-58,<br>01-2119956812-31,<br>01-0000000316-81 | Skin Irrit. 2; H315<br>Aquatic Acute 1;<br>H400<br>Aquatic Chronic 1;<br>H410<br><br>M-Factor (Acute aquatic toxicity): 1 | >= 2,5 - < 10            |
| α,β,2,2,3-pentamethylcyclopent-3-ene-1-butanol                                                                                      | 65113-99-7<br>265-453-0<br>01-2119975588-15                                                            | Aquatic Chronic 2;<br>H411<br>Eye Irrit. 2; H319                                                                          | >= 1 - < 2,5             |
| RM of (3E)-3-methyl-4-(2,6,6-trimethylcyclohex-2-en-1-yl)but-3-en-2-one & (1E)-1-(2,6,6-trimethylcyclohex-2-en-1-yl)pent-1-en-3-one | 127-51-5<br>204-846-3                                                                                  | Aquatic Chronic 2;<br>H411<br>Skin Sens. 1B; H317<br><br>M-Factor (Acute aquatic toxicity): 1                             | >= 1 - < 2,5             |
| citronellol                                                                                                                         | 106-22-9<br>203-375-0<br>01-2119453995-23                                                              | Skin Irrit. 2; H315<br>Skin Sens. 1B; H317<br>Eye Irrit. 2; H319                                                          | >= 1 - < 10              |
| 2-phenylethanol                                                                                                                     | 60-12-8<br>200-456-2<br>01-2119963921-31                                                               | Eye Irrit. 2; H319<br>Acute Tox. 4; H302                                                                                  | >= 1 - < 10              |

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

Version 1.0      Revision Date: 20.07.2023      SDS Number: 300001111100      Date of last issue: -  
Date of first issue: 20.07.2023

|                                                                              |                                                            |                                                                                                                                                                                                             |               |
|------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| [1R-(1,4,4a,6,8a)]-octahydro-4,8a,9,9-tetramethyl-1,6-methano-1(2H)-naphthol | 5986-55-0<br>227-807-2<br>01-2120754357-48                 | Aquatic Chronic 2; H411                                                                                                                                                                                     | >= 1 - < 2,5  |
| geraniol                                                                     | 106-24-1<br>203-377-1<br>603-241-00-5<br>01-2119552430-49  | Skin Irrit. 2; H315<br>Eye Dam. 1; H318<br>Skin Sens. 1; H317                                                                                                                                               | >= 1 - < 3    |
| 2-ethyl-3-hydroxy-4-pyrone                                                   | 4940-11-8<br>225-582-5<br>01-2120758795-36                 | Acute Tox. 4; H302                                                                                                                                                                                          | >= 1 - < 10   |
| $\alpha$ -methyl-1,3-benzodioxole-5-propionaldehyde                          | 1205-17-0<br>214-881-6                                     | Skin Sens. 1B; H317<br>Aquatic Chronic 2; H411<br>Repr. 2; H361<br><br>M-Factor (Acute aquatic toxicity): 1                                                                                                 | >= 1 - < 2,5  |
| caryophyllene                                                                | 87-44-5<br>201-746-1                                       | Asp. Tox. 1; H304<br>Skin Sens. 1B; H317                                                                                                                                                                    | >= 0,1 - < 1  |
| pin-2(3)-ene                                                                 | 80-56-8<br>201-291-9<br>01-2119519223-49                   | Flam. Liq. 3; H226<br>Asp. Tox. 1; H304<br>Skin Irrit. 2; H315<br>Skin Sens. 1B; H317<br>Aquatic Acute 1; H400<br>Aquatic Chronic 1; H410<br>Acute Tox. 4; H302<br><br>M-Factor (Acute aquatic toxicity): 1 | >= 0,25 - < 1 |
| (R)-p-mentha-1,8-diene                                                       | 5989-27-5<br>227-813-5<br>601-029-00-7<br>01-2119529223-47 | Skin Irrit. 2; H315<br>Skin Sens. 1B; H317<br>Aquatic Chronic 3; H412<br>Flam. Liq. 3; H226<br>Asp. Tox. 1; H304<br>Aquatic Acute 1; H400<br><br>M-Factor (Acute aquatic toxicity): 1                       | >= 0,25 - < 1 |
| (-)-pin-2(10)-ene                                                            | 127-91-3<br>242-060-2<br>01-2119519230-54                  | Asp. Tox. 1; H304<br>Skin Irrit. 2; H315<br>Flam. Liq. 3; H226<br>Skin Sens. 1B; H317<br>Aquatic Acute 1; H400<br>Aquatic Chronic 1; H410<br><br>M-Factor (Acute aquatic toxicity): 1                       | >= 0,25 - < 1 |

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

Version 1.0      Revision Date: 20.07.2023      SDS Number: 300001111100      Date of last issue: -  
Date of first issue: 20.07.2023

|                                                          |                                                                |                                                                                                                                                                                                                                          |                      |
|----------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| tetrahydro-4-methyl-2-(2-methylprop-1-enyl)pyran         | 16409-43-1<br>240-457-5<br>01-2119976300-42                    | Skin Irrit. 2; H315<br>Eye Irrit. 2; H319<br>Repr. 2; H361f                                                                                                                                                                              | $\geq 0,1 - < 1$     |
| 2,6-di-tert-butyl-p-cresol                               | 128-37-0<br>204-881-4<br>01-2119565113-46,<br>01-2119555270-46 | Aquatic Chronic 1;<br>H410<br>Aquatic Acute 1;<br>H400                                                                                                                                                                                   | $\geq 0,25 - < 1$    |
| 3,7,7-trimethylbicyclo[4.1.0]hept-3-ene                  | 13466-78-9<br>236-719-3                                        | Flam. Liq. 3; H226<br>Asp. Tox. 1; H304<br>Skin Sens. 1; H317<br>Skin Irrit. 2; H315<br>Aquatic Chronic 1;<br>H410                                                                                                                       | $\geq 0,1 - < 0,25$  |
| 1-(2,6,6-trimethyl-1,3-cyclohexadien-1-yl)-2-buten-1-one | 23696-85-7<br>245-833-2                                        | Skin Sens. 1A; H317<br>Aquatic Chronic 2;<br>H411<br>Skin Irrit. 2; H315<br><br>Acute toxicity estimate<br><br>Acute dermal toxicity:<br>2.900 mg/kg                                                                                     | $\geq 0,025 - < 0,1$ |
| 2,3-dihydro-2,2,6-trimethylbenzaldehyde                  | 116-26-7<br>204-133-7<br>01-2120116439-56-                     | Skin Irrit. 2; H315<br>Eye Irrit. 2; H319<br>Skin Sens. 1A; H317<br>Acute Tox. 4; H302<br>Aquatic Chronic 2;<br>H411<br><br>M-Factor (Acute aquatic toxicity): 1<br><br>Acute toxicity estimate<br><br>Acute oral toxicity:<br>500 mg/kg | $\geq 0,025 - < 0,1$ |

For explanation of abbreviations see section 16.

## SECTION 4: First aid measures

### 4.1 Description of first aid measures

- General advice : Take Hazard and Precautionary phrases (section 2) into account.
- If inhaled : Remove from exposure site to fresh air and keep at rest. If victim is unconscious, remove foreign bodies from the mouth. If victim has stopped breathing, give artificial respiration. Obtain medical advice.
- In case of skin contact : Remove contaminated clothes. Wash thoroughly with water

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



**777**

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

(and soap). Contact physician if symptoms persist.

In case of eye contact : Flush immediately with water for at least 15 minutes. Contact physician if symptoms persist.

If swallowed : Rinse mouth with water and obtain medical advice.

## 4.2 Most important symptoms and effects, both acute and delayed

Risks : Causes skin irritation.  
May cause an allergic skin reaction.  
Causes serious eye irritation.

## 4.3 Indication of any immediate medical attention and special treatment needed

## SECTION 5: Firefighting measures

### 5.1 Extinguishing media

Suitable extinguishing media : Carbondioxide, dry chemical, foam.

Unsuitable extinguishing media : Do not use a direct waterjet on burning material.

### 5.2 Special hazards arising from the substance or mixture

Specific hazards during fire-fighting : Water may be ineffective.

### 5.3 Advice for firefighters

Special protective equipment for firefighters : Wear self-contained breathing apparatus for firefighting if necessary.

Further information : Standard procedure for chemical fires.

## SECTION 6: Accidental release measures

### 6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Avoid inhalation and contact with skin and eyes. A self-contained breathing apparatus is recommended in case of a major spill.  
Prevent spreading over a wide area (e.g. by containment or oil barriers).

### 6.2 Environmental precautions

Environmental precautions : Keep away from drains, surface- and groundwater and soil.

### 6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Clean up spillage promptly. Remove ignition sources. Provide adequate ventilation. Avoid excessive inhalation of vapours.

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

Gross spillages should be contained by use of sand or inert powder and disposed of according to the local regulations.

## 6.4 Reference to other sections

## SECTION 7: Handling and storage

### 7.1 Precautions for safe handling

Advice on safe handling : Avoid excessive inhalation of concentrated vapors. Follow good manufacturing practices for housekeeping and personal hygiene. Wash any exposed skin immediately after any chemical contact, before breaks and meals, and at the end of each work period. Contaminated clothing and shoes should be thoroughly cleaned before re-use.

If appropriate, procedures used during the handling of this material should also be used when cleaning equipment or removing residual chemicals from tanks or other containers, especially when steam or hot water is used, as this may increase vapor concentrations in the workplace air. Where chemicals are openly handled, access should be restricted to properly trained employees.  
Keep all heated processes at the lowest necessary temperature in order to minimize emissions of volatile chemicals into the air.

Advice on protection against fire and explosion : Keep away from ignition sources and naked flame.

### 7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Store in a cool, dry, ventilated area away from heat sources.  
Keep containers upright and tightly closed when not in use.

### 7.3 Specific end use(s)

Specific use(s) : No information available.

## SECTION 8: Exposure controls/personal protection

### 8.1 Control parameters

#### Occupational Exposure Limits

|                                                 |          | Value type (Form of exposure) |                      | Basis  |
|-------------------------------------------------|----------|-------------------------------|----------------------|--------|
| 2,6-di-tert-butyl-p-cresol                      | 128-37-0 | VME                           | 10 mg/m <sup>3</sup> | FR VLE |
| Further information: Indicative exposure limits |          |                               |                      |        |

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006:

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

**777**

Version  
1.0

Revision Date:  
20.07.2023

SDS Number:  
300001111100

Date of last issue: -  
Date of first issue: 20.07.2023

|                                                                         |                                                                                           |              | Potential health effects   | Value                     |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------|----------------------------|---------------------------|
| 1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one | Workers                                                                                   | Skin contact | Long-term local effects    | 0,1011 mg/cm <sup>2</sup> |
| Remarks:                                                                | Exposure time: 8 h                                                                        |              |                            |                           |
|                                                                         | Workers                                                                                   | Skin contact | Long-term systemic effects | 1,73 mg/kg bw/day         |
| Remarks:                                                                | Exposure time: 8 h                                                                        |              |                            |                           |
|                                                                         | Workers                                                                                   | Inhalation   | Long-term systemic effects | 1,76 mg/m <sup>3</sup>    |
| Remarks:                                                                | Exposure time: 8 h                                                                        |              |                            |                           |
| (2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol        | Workers                                                                                   | Inhalation   | Long-term systemic effects | 21 mg/m <sup>3</sup>      |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | Workers                                                                                   | Dermal       | Long-term systemic effects | 6 mg/kg bw/day            |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | General population                                                                        | Inhalation   | Long-term systemic effects | 5,2 mg/m <sup>3</sup>     |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | General population                                                                        | Dermal       | Long-term systemic effects | 3 mg/kg bw/day            |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | General population                                                                        | Oral         | Long-term systemic effects | 3 mg/kg bw/day            |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
| geraniol                                                                | Workers                                                                                   | Inhalation   | Long-term systemic effects | 161,6 mg/m <sup>3</sup>   |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | Workers                                                                                   | Dermal       | Long-term systemic effects | 12,5 mg/kg bw/day         |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | Workers                                                                                   | Dermal       | Long-term local effects    | 11800 µg/cm <sup>2</sup>  |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | Workers                                                                                   | Dermal       | Acute local effects        |                           |
| Remarks:                                                                | Available hazard data do not enable the derivation of a DNEL for dermal irritant effects. |              |                            |                           |
|                                                                         | General population                                                                        | Inhalation   | Long-term systemic effects | 47,8 mg/m <sup>3</sup>    |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | General population                                                                        | Dermal       | Long-term systemic effects | 7,5 mg/kg bw/day          |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | General population                                                                        | Dermal       | Long-term local effects    | 11800 µg/cm <sup>2</sup>  |
| Remarks:                                                                | REACH data                                                                                |              |                            |                           |
|                                                                         | General population                                                                        | Dermal       | Acute local effects        |                           |
| Remarks:                                                                | Available hazard data do not enable the derivation of a DNEL for dermal                   |              |                            |                           |



# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

**777**

Version 1.0      Revision Date: 20.07.2023      SDS Number: 300001111100      Date of last issue: -  
Date of first issue: 20.07.2023

|          |                                                                                        |             |                            |                    |
|----------|----------------------------------------------------------------------------------------|-------------|----------------------------|--------------------|
|          | irritant effects.                                                                      |             |                            |                    |
|          | General population                                                                     | Oral        | Long-term systemic effects | 13,75 mg/kg bw/day |
| Remarks: | REACH data                                                                             |             |                            |                    |
|          | General population                                                                     | Eye contact | Local effects              |                    |
| Remarks: | Available hazard data do not enable the derivation of a DNEL for eye irritant effects. |             |                            |                    |

## Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006:

| Substance name                                                          | Environmental Compartment | Value                         |
|-------------------------------------------------------------------------|---------------------------|-------------------------------|
| 1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one | Fresh water               | 0,0028 mg/l                   |
|                                                                         | Marine water              | 0,00028 mg/l                  |
|                                                                         | Fresh water sediment      | 3,73 mg/kg dry weight (d.w.)  |
|                                                                         | Marine sediment           | 0,75 mg/kg dry weight (d.w.)  |
|                                                                         | Soil                      | 0,705 mg/kg dry weight (d.w.) |

## 8.2 Exposure controls

### Engineering measures

Where appropriate, use closed systems to transfer and process this material.  
If appropriate, isolate mixing rooms and other areas where this material is used or openly handled. Maintain these areas under negative air pressure relative to the rest of the plant.

### Personal protective equipment

Eye protection : Use tight-fitting goggles, face shield or safety glasses with side shields if eye contact might occur.  
Equipment should conform to EN 166

#### Hand protection

Material : Nitrile rubber  
Break through time : > 60 min  
Glove thickness : 0,38 mm

Material : Nitrile rubber  
Break through time : > 10 min  
Glove thickness : 0,1 mm

Remarks : Avoid skin contact. Use chemically resistant gloves. The selected protective gloves have to satisfy the specifications of Regulation (EU) 2016/425 and the standard EN 374 derived from it. Be aware that in daily use the durability of a chemical resistant protective glove can be notably shorter than the break through time measured according to EN 374, due to the numerous outside influences (e.g. temperature). Please observe the instructions regarding permeability and break-through time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions un-

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

der which the product is used, such as the danger of cuts, abrasion, and the contact time. The choice of an appropriate glove does not only depend on its material but also on other quality features and is different from one producer to the other.

Skin and body protection : Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place.  
Personal protection through wearing a tightly closed chemical protection suit and a self-contained breathing apparatus.

Respiratory protection : Use local exhaust ventilation around open tanks and other open sources of potential exposures in order to avoid excessive inhalation, including places where this material is openly weighed or measured. In addition, use general dilution ventilation of the work area to eliminate or reduce possible worker exposures.  
No respiratory protection is required during normal operations in a workplace where engineering controls such as adequate ventilation, etc. are sufficient.

If engineering controls and safe work practices are not sufficient, an approved, properly fitted respirator with organic vapor cartridges or canisters and particulate filters should be used:

- a) while engineering controls and appropriate safe work practices and/or procedures are being implemented; or
- b) during short term maintenance procedures when engineering controls are not in normal operation or are not sufficient; or
- c) if normal operational workplace vapor concentration in the air is increased due to heat ;
- d) during emergencies; or
- e) if engineering controls and operational practices are not sufficient to reduce airborne concentrations below an established occupational exposure limit.

Protective measures : To the extent deemed appropriate, implement pre-placement and regularly scheduled ascertainment of symptoms and spirometry testing of lung function for workers who are regularly exposed to this material.  
To the extent deemed appropriate, use an experienced air sampling expert to identify and measure volatile chemicals that could be present in the workplace air to determine potential exposures and to ensure the continuing effectiveness of engineering controls and operational practices to minimize exposure.

## SECTION 9: Physical and chemical properties

### 9.1 Information on basic physical and chemical properties

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



**777**

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

|                                                  |                               |
|--------------------------------------------------|-------------------------------|
| Physical state                                   | : liquid                      |
| Odour                                            | : conforms to standard        |
| Odour Threshold                                  | : not determined              |
| Melting point                                    | : not determined              |
| Boiling point                                    | : not determined              |
| Flammability                                     | : not determined              |
| Upper explosion limit / Upper flammability limit | : not determined              |
| Lower explosion limit / Lower flammability limit | : not determined              |
| Flash point                                      | : 84 °C<br>Method: closed cup |
| Auto-ignition temperature                        | : not determined              |
| Decomposition temperature                        | : not determined              |
| pH                                               | : not determined              |
| Viscosity, dynamic                               | : not determined              |
| Viscosity, kinematic                             | : not determined              |
| Water solubility                                 | : not determined              |
| Solubility in other solvents                     | : not determined              |
| Partition coefficient: n-octanol/water           | : not determined              |
| Vapour pressure                                  | : 0,66 hPa<br>Calculated      |
| Relative density                                 | : not determined              |
| Density                                          | : not determined              |

## 9.2 Other information

No data available

## SECTION 10: Stability and reactivity

### 10.1 Reactivity

No hazards to be specially mentioned.

### 10.2 Chemical stability

Stable under normal conditions.

### 10.3 Possibility of hazardous reactions

|                     |                                                                                  |
|---------------------|----------------------------------------------------------------------------------|
| Hazardous reactions | : Presents no significant reactivity hazard, by itself or in contact with water. |
|---------------------|----------------------------------------------------------------------------------|

### 10.4 Conditions to avoid

|                     |                           |
|---------------------|---------------------------|
| Conditions to avoid | : Direct sources of heat. |
|---------------------|---------------------------|

### 10.5 Incompatible materials

|                    |                                                                |
|--------------------|----------------------------------------------------------------|
| Materials to avoid | : Avoid contact with strong acids, alkali or oxidizing agents. |
|--------------------|----------------------------------------------------------------|

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

## 10.6 Hazardous decomposition products

Carbon monoxide and unidentified organic compounds may be formed during combustion.

## SECTION 11: Toxicological information

### 11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Acute toxicity

Not classified based on available information.

#### Product:

Acute oral toxicity : Acute toxicity estimate: > 2.000 mg/kg  
Method: Calculation method

#### Skin corrosion/irritation

Causes skin irritation.

#### Components:

##### 1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Species : reconstructed human epidermis (RhE)  
Assessment : Causes skin irritation.  
Method : OECD Test Guideline 439  
Result : irritating  
GLP : yes  
Test substance : (undiluted)  
Remarks : REACH data

##### (2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol:

Species : Rabbit  
Exposure time : 4 h  
Method : OECD Test Guideline 404  
Result : No skin irritation  
GLP : yes  
Remarks : REACH data

##### 1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:

Species : Human  
Exposure time : 0,25 h  
Assessment : Causes skin irritation.  
Method : EPISKIN Human Skin Model Test  
Result : irritating  
GLP : yes  
Test substance : (undiluted)  
Remarks : REACH data

Species : Human  
Method : OECD Test Guideline 431  
Result : No skin irritation  
GLP : yes  
Test substance : (undiluted)  
Remarks : REACH data

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

---

## Serious eye damage/eye irritation

Causes serious eye irritation.

### Components:

#### **geraniol:**

|               |                              |
|---------------|------------------------------|
| Species       | : Rabbit                     |
| Exposure time | : 1 h                        |
| Assessment    | : Causes serious eye damage. |
| Method        | : OECD Test Guideline 405    |
| Result        | : Severe eye irritation      |
| GLP           | : no                         |
| Remarks       | : REACH data                 |

#### **(2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol:**

|               |                           |
|---------------|---------------------------|
| Species       | : Rabbit                  |
| Exposure time | : 15 min                  |
| Method        | : OECD Test Guideline 405 |
| Result        | : Eye irritation          |
| GLP           | : yes                     |
| Remarks       | : REACH data              |

#### **1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:**

|            |                                  |
|------------|----------------------------------|
| Species    | : Chicken eye                    |
| Assessment | : Causes serious eye irritation. |
| Method     | : OECD 438                       |
| Result     | : Eye irritation                 |
| GLP        | : yes                            |
| Remarks    | : REACH data                     |

## Respiratory or skin sensitisation

### **Skin sensitisation**

May cause an allergic skin reaction.

### **Respiratory sensitisation**

Not classified based on available information.

### Components:

#### **1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:**

|            |                                                      |
|------------|------------------------------------------------------|
| Test Type  | : Local lymph node assay (LLNA)                      |
| Species    | : Mouse                                              |
| Assessment | : The product is a skin sensitiser, sub-category 1B. |
| Method     | : OECD Test Guideline 429                            |
| Result     | : Causes sensitisation.                              |
| GLP        | : yes                                                |
| Remarks    | : REACH data                                         |

#### **(2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol:**

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

Test Type : maximisation study  
Species : human  
Result : Did not cause sensitisation on laboratory animals.  
Test substance : 5%

## 1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:

Test Type : LLNA  
Species : Mouse  
Method : OECD 429  
Result : The product is a skin sensitiser, sub-category 1B.  
Remarks : REACH data

Test Type : repeated insult patch test  
Species : Humans  
Result : Does not cause skin sensitisation.  
Test substance : 10.% in ethanol/DEP (75:25)  
Remarks : REACH data

Species : Not tested on animals  
Result : Does not cause respiratory sensitisation.  
Remarks : REACH data

## Germ cell mutagenicity

Not classified based on available information.

## Components:

### 1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:

Genotoxicity in vitro : Test Type: Microbial mutagenesis assay (Ames test)  
Test system: Salmonella typhimurium  
Metabolic activation: with and without metabolic activation  
Method: OECD Test Guideline 471  
Result: negative  
GLP: yes  
Remarks: REACH data

Test Type: Microbial mutagenesis assay (Ames test)  
Test system: Salmonella typhimurium  
Metabolic activation: with and without metabolic activation  
Method: OECD Test Guideline 471  
Result: negative  
GLP: yes  
Remarks: REACH data

Test Type: Chromosome aberration test in vitro  
Test system: Human lymphocytes  
Metabolic activation: with and without metabolic activation  
Method: OECD 473  
Result: negative  
GLP: yes  
Remarks: REACH data

Test Type: In vitro mammalian cell gene mutation test

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

Test system: mouse lymphoma cells  
Metabolic activation: with and without metabolic activation  
Method: OECD Test Guideline 476  
Result: negative  
GLP: yes  
Remarks: REACH data

Genotoxicity in vivo : Test Type: In vivo micronucleus test  
Species: Mouse (male and female)  
Application Route: Dermal  
Method: OECD Test Guideline 474  
Result: negative  
GLP: yes

Test Type: In vivo micronucleus test  
Species: Rat (male and female)  
Application Route: Dermal  
Method: OECD Test Guideline 474  
Result: negative  
GLP: yes

## **(2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol:**

Genotoxicity in vitro : Test Type: In vitro mammalian cell gene mutation test  
Metabolic activation: with and without metabolic activation  
Method: OECD Test Guideline 476  
Result: negative  
GLP: yes  
Remarks: REACH data

## **1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:**

Genotoxicity in vitro : Test Type: Ames test  
Test system: Salmonella typhimurium  
Metabolic activation: with and without metabolic activation  
Method: OECD Test Guideline 471  
Result: negative  
GLP: yes  
Remarks: REACH data

Test Type: In vitro mammalian cell gene mutation test  
Test system: mouse lymphoma cells  
Metabolic activation: with and without metabolic activation  
Method: OECD Test Guideline 476  
Result: negative  
GLP: yes  
Remarks: REACH data

Test Type: in vitro mammalian cell micronucleus  
Test system: Human lymphocytes  
Metabolic activation: with and without metabolic activation  
Method: OECD Test Guideline 473  
Result: negative  
GLP: No information available.  
Remarks: REACH data

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

---

## Carcinogenicity

Not classified based on available information.

### Components:

#### **1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:**

|         |   |                            |
|---------|---|----------------------------|
| Species | : | Rat                        |
| Method  | : | see user defined free text |
| Remarks | : | not required<br>REACH data |

## Reproductive toxicity

Not classified based on available information.

### Components:

#### **1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:**

|                      |   |                                                                                                                                                                                                                                                                                                                                  |
|----------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effects on fertility | : | Test Type: reproductive and developmental toxicity study<br>Species: Rat, male and female<br>Strain: wistar<br>Application Route: Oral<br>General Toxicity - Parent: NOAEL: $\geq 300$ mg/kg body weight<br>Fertility: NOAEL: $\geq 300$ mg/kg body weight<br>Method: OECD Test Guideline 443<br>GLP: yes<br>Remarks: REACH data |
|----------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Test Type: reproductive and developmental toxicity study  
Species: Rat, male and female  
Strain: wistar  
Application Route: Oral  
General Toxicity - Parent: NOAEL: 120 mg/kg body weight  
Fertility: NOAEL:  $\geq 500$  mg/kg body weight  
Target Organs: Kidney, Liver, spleen  
Method: OECD Test Guideline 421  
GLP: yes  
Remarks: REACH data

|                               |   |                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Effects on foetal development | : | Test Type: Pre-natal<br>Species: Rat, female<br>Application Route: Oral<br>Duration of Single Treatment: 21 d<br>Frequency of Treatment: 1 daily<br>General Toxicity Maternal: NOAEL: 240 mg/kg body weight<br>Developmental Toxicity: NOAEL: $\geq 480$ mg/kg body weight<br>Method: OECD Test Guideline 414<br>GLP: yes<br>Remarks: REACH data |
|-------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Test Type: Pre-natal  
Species: Rabbit, female  
Application Route: Oral



# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

Duration of Single Treatment: 23 d  
Frequency of Treatment: 1 daily  
General Toxicity Maternal: NOAEL: 200 mg/kg body weight  
Developmental Toxicity: NOAEL: >= 500 mg/kg body weight  
Method: OECD Test Guideline 414  
GLP: yes  
Remarks: REACH data

## **(2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol:**

Effects on fertility : Test Type: reproductive and developmental toxicity study  
Species: Rat, male and female  
Application Route: Oral  
Dose: 1000 parts per million  
Duration of Single Treatment: 7 d  
Frequency of Treatment: 1 daily  
Method: OECD Test Guideline 422  
Result: No effects on fertility and early embryonic development were detected.  
GLP: yes  
Remarks: REACH data

Effects on foetal development : Test Type: reproductive and developmental toxicity study  
Species: Rat  
Application Route: Oral  
Dose: 300 milligram per kilogram  
Duration of Single Treatment: 14 d  
Method: OECD Test Guideline 414  
Result: No effects on fertility and early embryonic development were detected.  
GLP: yes  
Remarks: REACH data

## **1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:**

Effects on fertility : Test Type: reproductive and developmental toxicity study  
Species: Rat, male and female  
Strain: wistar  
Application Route: Ingestion  
General Toxicity - Parent: NOAEL: 150 mg/kg food  
Fertility: NOAEL: 1,875 mg/kg food  
Early Embryonic Development: NOAEL: 1,875 mg/kg food  
Symptoms: No effects on fertility, No effects on foetal development  
Target Organs: Kidney, Liver  
Method: OECD Test Guideline 421  
GLP: yes  
Remarks: REACH data

Effects on foetal development : Test Type: reproductive and developmental toxicity study  
Species: Rat, female  
Strain: wistar  
Application Route: Ingestion  
Developmental Toxicity: NOAEL: 121,83 mg/kg body weight  
Symptoms: No effects on newborn

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

Target Organs: Kidney, Liver  
Method: OECD Test Guideline 421  
Result: No effects on fertility and early embryonic development were detected.  
GLP: yes  
Remarks: REACH data

## STOT - single exposure

Not classified based on available information.

## STOT - repeated exposure

Not classified based on available information.

## Repeated dose toxicity

### Components:

#### **1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:**

Species : Rat, male and female  
NOAEL : 150 mg/kg  
Application Route : Oral  
Exposure time : 28-day  
Number of exposures : 1x /day  
Method : OECD Test Guideline 407  
GLP : yes  
Remarks : REACH data

Species : Rat, male and female  
NOAEL : 120 mg/kg  
Application Route : Oral  
Exposure time : 90-day  
Number of exposures : 1x /day  
Method : OECD Test Guideline 408  
GLP : yes  
Target Organs : spleen, Liver, Kidney  
Remarks : REACH data

Species : Rat, male and female  
NOAEL : 250 mg/kg  
Application Route : Dermal  
Exposure time : 90-day  
Method : OECD Test Guideline 411  
GLP : yes  
Target Organs : Skin, Liver  
Remarks : REACH data

Species : Mouse, male and female  
NOAEL : 500 mg/kg  
Application Route : Dermal  
Exposure time : 90-day  
Method : OECD Test Guideline 411  
GLP : yes  
Target Organs : Skin, Liver  
Remarks : REACH data

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

**777**

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

## **(2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol:**

|                   |                           |
|-------------------|---------------------------|
| Species           | : Rat, male               |
| NOAEL             | : 981 mg/kg               |
| NOAEL             | : 981 mg/kg               |
| Application Route | : Oral                    |
| Exposure time     | : 90-day                  |
| Method            | : OECD Test Guideline 408 |
| GLP               | : yes                     |
| Remarks           | : REACH data              |

|                   |                           |
|-------------------|---------------------------|
| Species           | : Rat, female             |
| NOAEL             | : 1109 mg/kg              |
| NOAEL             | : 1.109 mg/kg             |
| Application Route | : Oral                    |
| Exposure time     | : 90-day                  |
| Method            | : OECD Test Guideline 408 |
| GLP               | : yes                     |
| Remarks           | : REACH data              |

## **1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:**

|                     |                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------|
| Species             | : Rat, male and female                                                                             |
| NOAEL               | : 10 mg/kg                                                                                         |
| Application Route   | : Oral                                                                                             |
| Exposure time       | : 90-day                                                                                           |
| Number of exposures | : 1x /day                                                                                          |
| Method              | : OECD Test Guideline 408                                                                          |
| GLP                 | : yes                                                                                              |
| Target Organs       | : Kidney                                                                                           |
| Assessment          | : The substance or mixture is not classified as specific target organ toxicant, repeated exposure. |
| Remarks             | : REACH data                                                                                       |

### **Aspiration toxicity**

Not classified based on available information.

### **Components:**

#### **pin-2(3)-ene:**

May be fatal if swallowed and enters airways.

## **11.2 Information on other hazards**

### **Endocrine disrupting properties**

#### **Product:**

|            |                                                                                                                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assessment | : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher. |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

## SECTION 12: Ecological information

### 12.1 Toxicity

#### Components:

##### **1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:**

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 1,3 mg/l  
Exposure time: 96 h  
Test Type: semi-static test  
Analytical monitoring: yes  
Method: OECD Test Guideline 203  
GLP: yes  
Remarks: REACH data

LC50 (Lepomis macrochirus (Bluegill sunfish)): 1,8 mg/l  
Exposure time: 48 h  
Test Type: semi-static test  
Analytical monitoring: yes  
Method: OECD Test Guideline 203  
GLP: yes  
Remarks: REACH data

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 1,38 mg/l  
Exposure time: 48 h  
Test Type: semi-static test  
Analytical monitoring: yes  
Method: OECD Test Guideline 202  
GLP: yes  
Remarks: REACH data

Toxicity to algae/aquatic plants : ErC50 (Desmodesmus subspicatus (green algae)): > 2,6 mg/l  
End point: Growth inhibition  
Exposure time: 72 h  
Test Type: static test  
Analytical monitoring: yes  
Method: OECD Test Guideline 201  
GLP: yes  
Remarks: REACH data

NOEC (Desmodesmus subspicatus (green algae)): >= 2,6 mg/l  
End point: Growth inhibition  
Exposure time: 72 h  
Analytical monitoring: yes  
Method: OECD Test Guideline 201  
GLP: yes  
Remarks: REACH data

Toxicity to microorganisms : NOEC (activated sludge): > 100 mg/l  
Exposure time: 42 d  
Test Type: static test  
Method: OECD 301 F

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

GLP: yes  
Remarks: REACH data

Toxicity to fish (Chronic toxicity) : NOEC: 0,16 mg/l  
Exposure time: 30 d  
Species: Danio rerio (zebra fish)  
Analytical monitoring: yes  
Method: OECD Test Guideline 210  
GLP: yes  
Remarks: REACH data

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 0,028 mg/l  
End point: Reproduction  
Exposure time: 21 d  
Species: Daphnia magna (Water flea)  
Analytical monitoring: yes  
Method: OECD Test Guideline 211  
GLP: yes  
Remarks: REACH data

NOEC: 0,448 mg/l  
End point: mortality  
Exposure time: 21 d  
Species: Daphnia magna (Water flea)  
Analytical monitoring: yes  
Method: OECD Test Guideline 211  
GLP: yes  
Remarks: REACH data

EC10: 0,044 mg/l  
End point: Reproduction  
Exposure time: 21 d  
Species: Daphnia magna (Water flea)  
Analytical monitoring: yes  
Method: OECD Test Guideline 211  
GLP: yes  
Remarks: REACH data

## Ecotoxicology Assessment

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

## 12.2 Persistence and degradability

### Components:

#### **1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:**

Biodegradability : Test Type: aerobic  
Inoculum: activated sludge, non-adapted  
Concentration: 18,8 mg/l  
Result: Readily biodegradable.  
Biodegradation: 96,3 %  
Exposure time: 28 d  
Method: OECD Test Guideline 301F

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

GLP: yes  
Remarks: REACH data

## **(2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol:**

Biodegradability : Concentration: 2 mg/l  
Result: Not readily biodegradable.  
Biodegradation: 5 %  
Exposure time: 28 d  
Method: OECD Test Guideline 301D  
GLP: yes  
Remarks: REACH data

## **1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:**

Biodegradability : Test Type: aerobic  
Inoculum: activated sludge  
Concentration: 100 mg/l  
Result: Not readily biodegradable.  
Biodegradation: 0 %  
Related to: Biochemical oxygen demand  
Exposure time: 28 d  
Method: OECD Test Guideline 301C  
GLP: yes  
Remarks: REACH data

## **12.3 Bioaccumulative potential**

### **Components:**

#### **1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:**

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)  
Exposure time: 21 d  
Bioconcentration factor (BCF): 391  
Method: OECD Test Guideline 305  
GLP: yes  
Remarks: REACH data

#### **(2E)-2-ethyl-4-(2,2,3-trimethylcyclopent-3-en-1-yl)but-2-en-1-ol:**

Bioaccumulation : Bioconcentration factor (BCF): 1.096,5  
Method: Calculated  
GLP: no  
Remarks: REACH data

Partition coefficient: n-octanol/water : log Pow: 4,4 (35 °C)  
Method: OECD Test Guideline 117  
GLP: yes  
Remarks: REACH data

#### **1,2,3,5,6,7-hexahydro-1,1,2,3,3-pentamethyl-4H-inden-4-one:**

Bioaccumulation : Species: Cyprinus carpio (Carp)  
Exposure time: 60 d  
Bioconcentration factor (BCF): 157

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



**777**

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

Method: OECD Test Guideline 305  
GLP: yes  
Remarks: REACH data

Partition coefficient: n-octanol/water : log Pow: 4,2 (20 °C)  
Method: OECD 117  
GLP: yes  
Remarks: REACH data

## 12.4 Mobility in soil

### Product:

Mobility : Remarks: No data available

### Components:

#### **1-(1,2,3,4,5,6,7,8-Octahydro-2,3,8,8-tetramethyl-2-naphthyl)ethan-1-one:**

Distribution among environmental compartments : Adsorption/Soil  
Koc: 12598, log Koc: 4,1

## 12.5 Results of PBT and vPvB assessment

### Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

## 12.6 Endocrine disrupting properties

### Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

## 12.7 Other adverse effects

### Product:

Additional ecological information : There is no data available for this product.

---

## SECTION 13: Disposal considerations

### 13.1 Waste treatment methods

Product : Dispose of according to local regulations. Avoid disposing into drainage systems and into the environment.

Contaminated packaging : Empty containers should be taken to an approved waste han-

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

ding site for recycling or disposal.

## SECTION 14: Transport information

### 14.1 UN number or ID number

|      |   |         |
|------|---|---------|
| ADR  | : | UN 3082 |
| IMDG | : | UN 3082 |
| IATA | : | UN 3082 |

### 14.2 UN proper shipping name

|      |   |                                                                                                         |
|------|---|---------------------------------------------------------------------------------------------------------|
| ADR  | : | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,<br>N.O.S.<br>(Octahydro Tetramethyl Naphthalenyl Ethanone) |
| IMDG | : | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,<br>N.O.S.<br>(Octahydro Tetramethyl Naphthalenyl Ethanone) |
| IATA | : | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,<br>N.O.S.<br>(Octahydro Tetramethyl Naphthalenyl Ethanone) |

### 14.3 Transport hazard class(es)

|      |   |   |
|------|---|---|
| ADR  | : | 9 |
| IMDG | : | 9 |
| IATA | : | 9 |

### 14.4 Packing group

|                                          |                                 |
|------------------------------------------|---------------------------------|
| <b>ADR</b>                               |                                 |
| Packing group                            | : III                           |
| Classification Code                      | : M6                            |
| Hazard Identification Number             | : 90                            |
| Labels                                   | : 9                             |
| Tunnel restriction code                  | : (-)                           |
| <b>IMDG</b>                              |                                 |
| Packing group                            | : III                           |
| Labels                                   | : 9                             |
| EmS Code                                 | : F-A, S-F                      |
| <b>IATA (Cargo)</b>                      |                                 |
| Packing instruction (cargo aircraft)     | : 964                           |
| Packing instruction (LQ)                 | : Y964                          |
| Packing group                            | : III                           |
| Labels                                   | : Miscellaneous Dangerous Goods |
| <b>IATA_P (Passenger)</b>                |                                 |
| Packing instruction (passenger aircraft) | : 964                           |



# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

Packing instruction (LQ) : Y964  
Packing group : III  
Labels : Miscellaneous Dangerous Goods

## 14.5 Environmental hazards

### ADR

Environmentally hazardous : yes

### IMDG

Marine pollutant : yes (Octahydro Tetramethyl Naphthalenyl Ethanone)

### IATA (Passenger)

Environmentally hazardous : yes

### IATA (Cargo)

Environmentally hazardous : yes

## 14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

## 14.7 Maritime transport in bulk according to IMO instruments

Not applicable for product as supplied.

## SECTION 15: Regulatory information

### 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

REACH - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles (Annex XVII) : Conditions of restriction for the following entries should be considered: Number on list 3

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors : acetone

REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59). : Not applicable

Regulation (EC) No 1005/2009 on substances that deplete the ozone layer : Not applicable

Regulation (EU) 2019/1021 on persistent organic pollutants (recast) : Not applicable

REACH - List of substances subject to authorisation (Annex XIV) : Not applicable

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors

acetone (ANNEX II)

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



**777**

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

Reinforced medical supervision (R4624-18) : The product has no CMR properties

## Other regulations:

Take note of Directive 94/33/EC on the protection of young people at work or stricter national regulations, where applicable.

## 15.2 Chemical safety assessment

A Chemical Safety Assessment is not required for this substance.

## SECTION 16: Other information

### Full text of H-Statements

|       |                                                         |
|-------|---------------------------------------------------------|
| H226  | : Flammable liquid and vapour.                          |
| H302  | : Harmful if swallowed.                                 |
| H304  | : May be fatal if swallowed and enters airways.         |
| H315  | : Causes skin irritation.                               |
| H317  | : May cause an allergic skin reaction.                  |
| H318  | : Causes serious eye damage.                            |
| H319  | : Causes serious eye irritation.                        |
| H361  | : Suspected of damaging fertility or the unborn child.  |
| H361f | : Suspected of damaging fertility.                      |
| H400  | : Very toxic to aquatic life.                           |
| H410  | : Very toxic to aquatic life with long lasting effects. |
| H411  | : Toxic to aquatic life with long lasting effects.      |
| H412  | : Harmful to aquatic life with long lasting effects.    |

### Full text of other abbreviations

|                 |                                        |
|-----------------|----------------------------------------|
| Acute Tox.      | : Acute toxicity                       |
| Aquatic Acute   | : Short-term (acute) aquatic hazard    |
| Aquatic Chronic | : Long-term (chronic) aquatic hazard   |
| Asp. Tox.       | : Aspiration hazard                    |
| Eye Dam.        | : Serious eye damage                   |
| Eye Irrit.      | : Eye irritation                       |
| Flam. Liq.      | : Flammable liquids                    |
| Repr.           | : Reproductive toxicity                |
| Skin Irrit.     | : Skin irritation                      |
| Skin Sens.      | : Skin sensitisation                   |
| FR VLE          | : France. Occupational Exposure Limits |
| FR VLE / VME    | : Time Weighted Average                |

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration as-

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



777

|                |                              |                             |                                                          |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|
| Version<br>1.0 | Revision Date:<br>20.07.2023 | SDS Number:<br>300001111100 | Date of last issue: -<br>Date of first issue: 20.07.2023 |
|----------------|------------------------------|-----------------------------|----------------------------------------------------------|

sociated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

## Further information

Other information : In December 2003, the National Institute for Occupational Safety and Health ("NIOSH") published an Alert on preventing lung disease in workers who use or make flavorings [NIOSH Publication Number 2004-110].  
In August 2004, the United States Flavor and Extract Manufacturers Association (FEMA) issued a report entitled "Respiratory Safety in the Flavor Manufacturing Workplace".  
Both of these reports provide recommendations for reducing employee exposure and for medical surveillance in the workplace. The recommendations in these reports are generally applicable to the use of any chemical in the workplace and you are strongly urged to review both of these reports.  
The report published by FEMA also contains a list of "high priority" chemicals. If any of these chemicals are present in this product at a concentration  $\geq 1.0\%$  due to an intentional addition by IFF, the chemical(s) will be identified in this safety data sheet.

According to Regulation (EC) No. 1907/2006 the information in this safety data sheet is based on the properties of the material known to IFF at the time the data sheet was issued. The safety data sheet is intended to provide information for a health and safety assessment of the material and the circumstances, under which it is packaged, stored or applied in the workplace. For such a safety assessment International Flavors & Fragrances holds no responsibility. This document is not intended for quality assurance purposes.

# SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



**777**

|         |                |              |                                 |
|---------|----------------|--------------|---------------------------------|
| Version | Revision Date: | SDS Number:  | Date of last issue: -           |
| 1.0     | 20.07.2023     | 300001111100 | Date of first issue: 20.07.2023 |

---

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

FR / EN